



Age, Growth and Reproduction of Silvery Pout (*Gadiculus argenteus* Guichenot, 1850) in the Saros Bay (Northeastern Aegean Sea)

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Received: 19 February 2018 / Revised: 6 December 2018 / Published online: 18 February 2019
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Abstract

The age, growth and reproduction of Silvery Pout (*Gadiculus argenteus*), caught in the Bay of Saros (Northeastern Aegean Sea) between April 2005 and September 2008, were investigated. The total length and weight of the *G. argenteus* ranged from 5.5 cm to 16.8 cm in total length (TL) and from 1.6 g to 30.7 g in weight. The length-weight relationships were estimated $W = 0.011 \cdot TL^{2.924}$. Age data derived from otoliths readings were used to estimate the growth parameters of the von Bertalanffy equation. The population growth parameters for *G. argenteus* were estimated as $L_{\infty} = 14.97$ cm, $K = 0.28 \text{ year}^{-1}$ and $t_0 = -1.69$ year. The maximum age was 5 years. The data on maturity stages indicated that length at 50% maturity (L_{50}) were 8.94 cm TL for *G. argenteus*. Monthly values of the gonadosomatic index indicated that reproduction period of *G. argenteus* was determined between December and March.

Keywords Age · Growth · Reproduction · Silvery pout · *Gadiculus argenteus* · Aegean Sea

Introduction

The silvery pout (*Gadiculus argenteus* Guichenot, 1850) is a deep water, open sea epi-to mesopelagic fish common over the edge of the continental shelf. It occurs in large schools over mud, muddy sand, gravel and rock bottoms, from 110 to 1000 m depth, but mostly from 200 to 400 m in the northern Mediterranean and from 300 to 600 m off Tunisia (Cohen et al. 1990). It breeds from December to January in the Mediterranean, and from mid-winter to spring in northern Europe. The major spawning grounds are located in the western part of the Mediterranean, on both sides of the Straits of Gibraltar in the deep water. There is relatively little information on this species. This species is thought to spawn over much of its distribution range (Hislop et al. 2015). The inclusion of larvae in Munk and

Nielsen (2005) suggests that larvae at least occur in the North Sea, however, these may simply be advected in to the area from off shelf spawning grounds. There is not enough information to picture a spawning map for silvery pout. According to Cohen et al. (1990) occurs in mid winter and spring. However, the early spawning is probably confined to southwestern distribution area, i.e. west and south of Ireland. In the North Sea spawning is thought to occur between March and May (Munk and Nielsen 2005). Silvery pout rarely lives up to 3 years, when it reaches a maximum length (15cm) most commonly, it attains 13 cm in its northern distribution, and 7 to 10 cm in the Mediterranean Sea. Feeds on small crustaceans and possibly on worms. It is preyed upon by a number of larger valuable fish (Cohen et al. 1990). Silvery pout is of no particular economic importance and is usually bycatch in small-scale and artisanal bottom trawling fisheries off Spain and Sicily (Cohen et al. 1990).

There are a few studies on biology (Sotiropoulou 1999; Stergiou and Politou 1995; Stergiou and Karachle 2006; Bilge et al. 2015), length-weight relationships (Vassilopoulou 1990; Pereda and Villamor 1991; Borges et al. 2003; Lamprakis et al. 2003; Filiz and Bilge 2004; Torres et al. 2012; Wilhelms 2013), population structure and diet (Albert 1993; Fernandez-Arcaya et al. 2013; Keskin et al. 2014). Though the silvery pout is one important deep sea fishes but information on its biology is very limited. Little is known of the distribution, biology and ecology of this species, thus

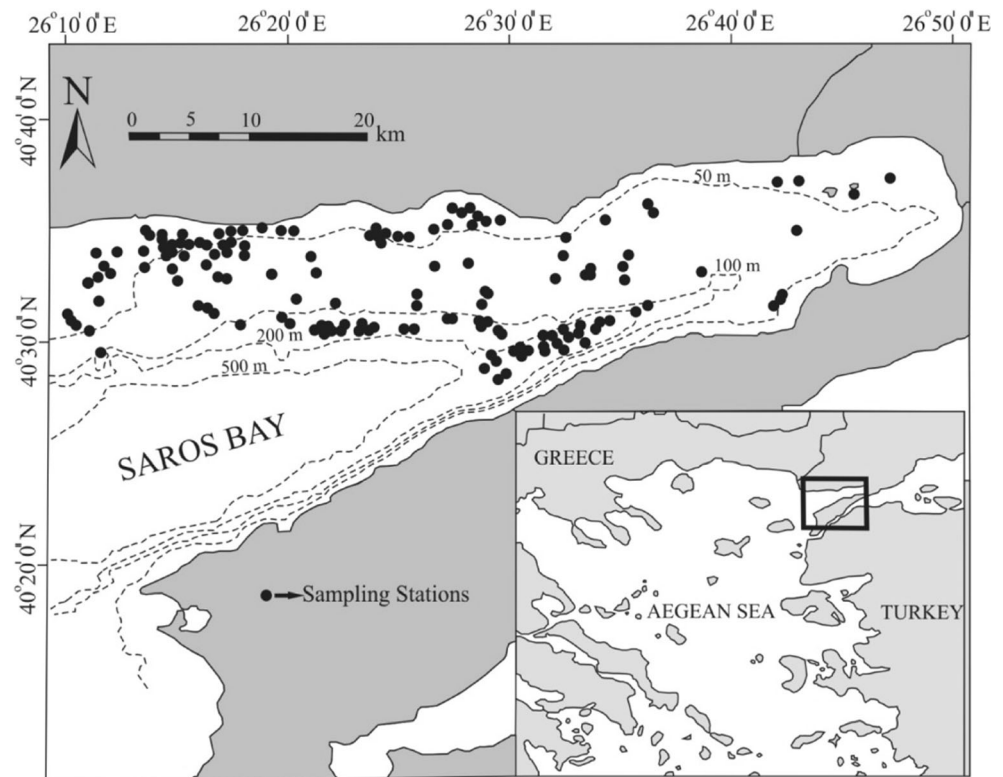
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Fig. 1 Trawl sampling stations in Saros Bay, the North Aegean Sea



describing the populations in some details is necessary. The aim of this paper is to describe the age, growth, reproduction and population structure of *G. argenteus* in Saros Bay, Northeastern Aegean Sea. The result of the present work will contribute to the knowledge about the age composition and growth of *G. argenteus* and also to a better understanding of its role the ecosystem.

Material and Methods

Samples of *G. argenteus* were collected between April 2005 and September 2008 from monthly samples at depths ranged between 100 and 500 m. from the Saros Bay (Fig. 1). At each station a bottom trawl net with a 44 mm stretched mesh size at the cod-end was towed for 30 min at approximately 2.5 knots.

The length-weight relationships were determined according to the allometric equation (Sparre et al. 1989): $W = aL^b$, where W is the total body weight (g), L is the total length (cm) while a and b are constants.

The age was determined using the otoliths. Sagittal otoliths were extracted through dissection of the otic bulla, otoliths were then cleaned, labeled and stored in plastic tubes (Bagenal and Tesch 1978; Bilge et al. 2015). Whole otoliths ($n = 310$) were examined with a stereomicroscope Olympus SZX16 to identify and count growth bands. Annual rings on the whole otolith were counted in glycerin.

Growth was expressed in terms of the von Bertalanffy equation (Beverton and Holt 1957): $L_t = L_\infty (1 - e^{-K(t-t_0)})$, where L_∞ is the asymptotic total length, L_t the total length at age t , K the growth curvature parameter and t_0 is the theoretical age when fish would have been at zero total length. Growth parameters were estimated according to the non-linear method by using the FISAT program package (Sparre et al. 1989). To compare with other studies, growth performance index was estimated (Pauly and Munro 1984). This test provided an indication of the reliability of age estimates, since it had been suggested that phi-prime test values were similar for the same species and genera. The test was based on: $\Phi' = \log K + 2 \log L_\infty$ (Pineiro and Sainza 2003). The sex and maturity stage of each specimen were determined by macroscopic examination of the gonads. Stages of maturation were classified according to the Holden

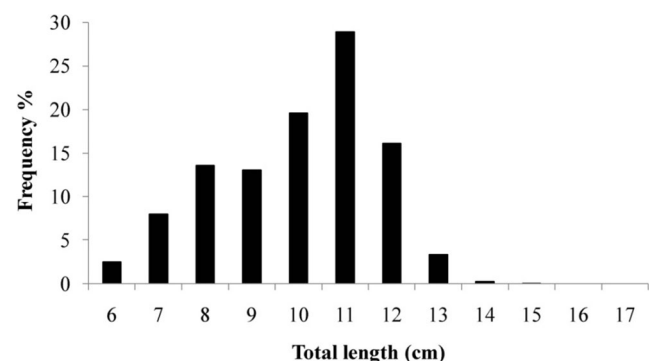


Fig. 2 Length-frequency distribution of *G. argenteus*

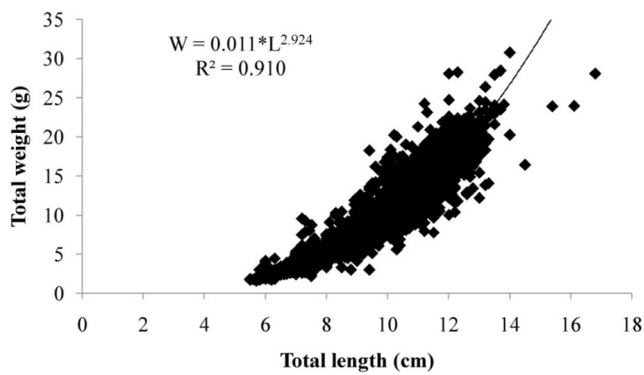


Fig. 3 Length-weight relationship of *G. argenteus*

and Raitt (1974) scale. Specimens were assigned to one of four categories: immature, maturing, mature and ripe. The gonadosomatic index (GSI) was calculated monthly by the equation: $GSI = (\text{gonad weight} / \text{fish weight without gonad}) * 100$. Size at first maturity (L_{50}) was defined as the size at which 50% of individuals were mature. Specimens were grouped in 1 cm size classes and the proportion of mature and immature individuals recorded (Cherif et al. 2007). The percentages of maturity by length class and sex were fitted to a logistic function using the Newton algorithm from @Microsoft excel solver routine: $P(1) = 1 / (1 + e^{-(a + bL)})$ where $P(1)$ was the proportion of mature fish at length 1, and a and b the parameters of the logistic equation (Pineiro and Sainza 2003).

Results and Discussion

A total of 3828 *G. argenteus* were sampled for this study. The length and weight of the *G. argenteus* ranged from 5.5 cm to 16.8 cm in total length (TL) and from 1.6 g to 30.7 g in weight, respectively. The length frequency distribution in the *G. argenteus* population showed that the most frequent size classes were 11 cm TL, accounting for 30% (Fig. 2).

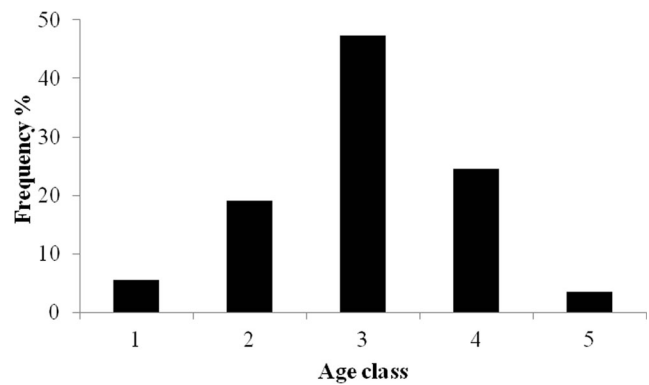


Fig. 4 Age frequency distribution of *G. argenteus*

Length-weight relationships were estimated $W = 0.011 * TL^{2.924}$, $r^2 = 0.91$, $n = 3828$ (Fig. 3). Value of the exponent b was 2.924, %95 CL for b is 2.89–2.95. ($r^2 = 0.91$) which indicated a negative allometric growth ($P < 0.05$).

Studies on the length-weight relationship (LWR) of fishes are an important tool to describe several biological aspects of the species. Consequently, these studies are relevant due the necessity to comprehend about fish life cycle (Freitas et al. 2017). A comparison of published length-weight relationships for *G. argenteus* is shown in Table 1. The allometric coefficient b varied from 2.72 to 3.49 for species in different regions. According to the parameter b , the species *G. argenteus* had an negative allometric growth pattern ($b < 3$) in this study. Generally, in the length-weight relationship studies, the growth pattern (b) within the same species can change depending on the season, food availability, population, sex, environmental conditions or physiology (Fulton 1904; Karakulak et al. 2006; Fontoura et al. 2010; Freitas et al. 2017).

Otoliths were selected for age determination of *G. argenteus*. The age was determined by counting the annual ring marks on the surface of the otoliths in 310 specimens. The maximum age of fish estimated was 5 years. The age 3 (47%) were dominant, followed by age groups 4 (25%), 2 (19%),

Table 1 Length-weight relationships of *G. argenteus* obtained by different authors

N	Sex*	Size range (cm)	a	b	r ²	Region	Author
317	Σ	3.5–15.0	0.0180	2.73	0.87	North Evvoikos Gulf, Aegean Sea	Vassilopoulou (1990)
66	–	4.0–13.0	0.0023	3.49	0.96	Cantabrian Sea, N Spain (NE Atlantic)	Pereda and Villamor (1991)
209	Σ	5.7–12.9	0.0048	3.09	0.82	G. N. Evvoikos and Trikeri Channel	Stergiou and Politou (1995)
235	–	7.0–14.6	0.0064	3.14	0.86	Algarve, Portugal (NE Atlantic)	Borges et al. (2003)
102	–	4.9–10.0	0.0088	2.97	0.84	North Aegean Sea (Thracian Sea)	Lamprakis et al. (2003)
110	–	6.4–10.5	0.0056	3.24	0.89	North Aegean Sea	Filiz and Bilge (2004)
156	Σ	5.3–14.0	0.0039	3.33	0.96	Gulf of Cadiz, SW Spain	Torres et al. (2012)
102	–	5.8–13.3	0.0154	2.72	0.89	North Sea	Wilhelms (2013)
286	Σ	4.8–12.1	0.0055	3.25	0.97	Southern Aegean Sea	Bilge et al. (2015)
3828	Σ	5.5–16.8	0.011	2.92	0.91	Saros Bay (North Aegean Sea)	This study

Sex*: “Σ” Male and Female; “–” unidentified in the text

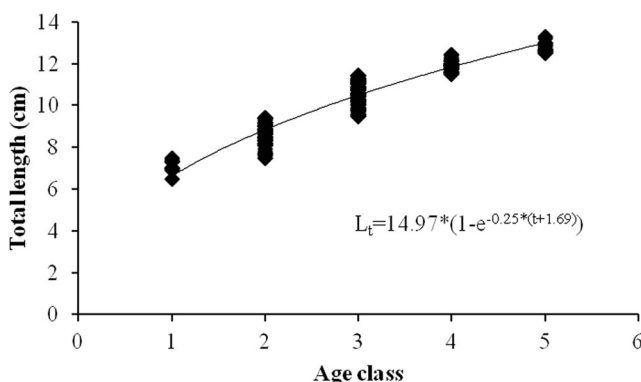
Table 2 Mean, minimum and maximum length at each age class of *G. argenteus*

Age	N	Length (cm)		
		Min.	Max.	Mean
1	10	6.5	7.5	7.1
2	76	7.5	9.4	8.6
3	146	9.5	11.4	10.6
4	59	11.5	12.4	11.9
5	17	12.5	13.3	12.8

1 (6%) and age group 5 (4%) (Fig. 4). The number of individuals and length interval at age data according to sexes combined are shown in Table 2. The von Bertalanffy population growth parameters for *G. argenteus* were estimated as $L_{\infty} = 14.97$ cm, $K = 0.28 \text{ year}^{-1}$ and $t_0 = -1.69$ year (Fig. 5).

The study results indicated that the age composition of *G. argenteus* ranged between 1 and 5 years. Age information is an essential component in growth rate, mortality rate and productivity estimations, hence it is classified among the most influential of biological variables (Campana 2014). *G. argenteus* is one important demersal fish species which is little known in Turkish waters, there is only one study about age and growth parameters on *G. argenteus* (Bilge et al. 2015). The calculated growth performance index (Φ') was 1.86 in Bilge et al. (2015) and 1.75 our study. In order to compare the growth of the *G. argenteus* population with others, all available literature data of von Bertalanffy growth parameters and ϕ values, including results from the present study, are reported (Table 3).

Relationships between otolith length (mm) and fish size (total length, cm) were described by means of linear regression analysis for silvery pout (*G. argenteus*) from the NE Aegean Sea (Fig. 6). Otoliths are usually used to identify fish species and to estimate their age and size, information that is useful for population dynamics and feeding habit studies (Viva et al. 2015). The estimation of body size from otolith measurement is possible by involving the regression between otolith length and fish size (Templemann and Squires 1956;

**Fig. 5** The von Bertalanffy growth curve for *G. argenteus***Table 3** Parameters of von Bertalanffy growth Eq. (K , L_{∞} , t_0) obtained by different authors

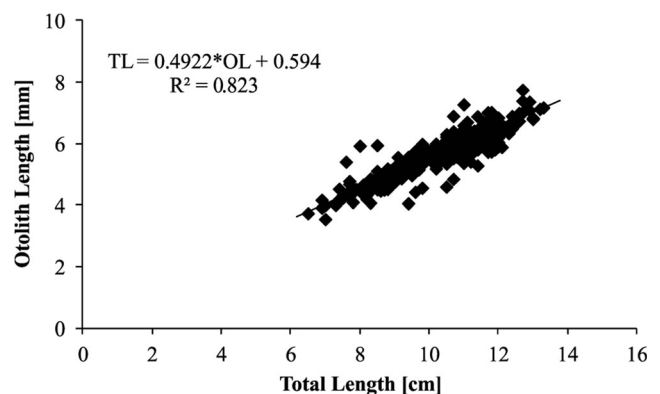
Author	Region	Sex	L_{∞}	K	t_0	Φ'
Vassilopoulou (1990)	Northern Euboean Gulf	–	19.7	0.19	–0.94	1.87
Stergiou and Politou (1995)	North Evvoikos Gulf	–	16.2	0.50	–	2.12
Sotiropoulou (1999)	Evvoikos Gulf	Σ	16.6	0.50	–	–
Sotiropoulou (1999)	Korinthiakos-Patraikos Gulfs	Σ	14.0	0.52	–	–
Sotiropoulou (1999)	Ionian Sea	Σ	13.5	0.50	–	–
Bilge et al. (2015)	Aegean Sea	Σ	13.26	0.42	–1.45	1.86
This study	Saros Bay	Σ	14.97	0.25	–1.69	1.75

Sex: “Σ” Male and Female; “–” unsexed; Φ' (growth performance index): similarity index with other studies

Echeverria 1987; Panfili et al. 2002). Only one study about otolith relative size and function related morphologies of *G. argenteus* at 300–450 m from the north-western Mediterranean Sea (Lombarte and Cruz 2007). The present study is first reference on the relationship between otolith length and fish size for the silvery pout species investigated in the Saros Bay (NE Aegean Sea).

The gonadosomatic index (GSI) was used to determine the reproductive period, which was calculated from samples taken monthly. GSI analysis showed that the spawning period began in December and continued until March, reaching the maximum average GSI value in January (9.35%) and ranged from 0.21% to 9.35% (Fig. 7).

In this study, due to the small number of males, only females maturity was determined. The reproduction period of *G. argenteus* females were found to be between December and March. The maximum average GSI value was obtained in January. Similar results were reported from Whitehead et al. (1986) and Cohen et al. (1990) who defined the reproductive period of *G. argenteus* in the Mediterranean Sea extended from December to

**Fig. 6** Relationships between otolith length-total length of *G. argenteus*

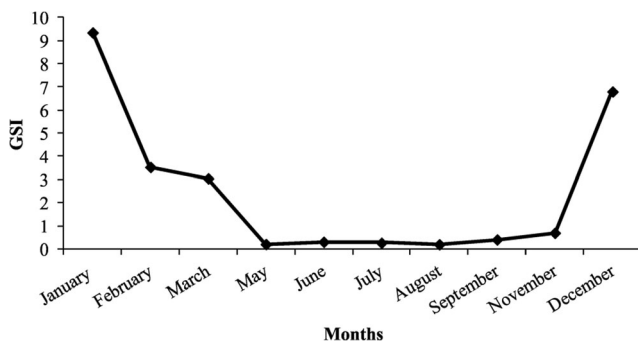


Fig. 7 The gonadosomatic index values (GSI) of *G. argenteus*

January. Another study based on larvae abundance was from Sartini and Belcari (2003) who found that minimum and maximum larvae abundance of *G. argenteus argenteus* in March, between 50 and 200 m depth in the Northern Tyrrhenian Sea.

Percentage of each gonadal development stage is indicated in Fig. 8. According to the gonadic maturation stages, 1.15% of the total fishes were immature (I), 54.5% were maturing (II), 5.7% were mature (III), and 38.5% were ripe (IV). Therefore, 44.2% of the fish were in the reproductive process. The highest proportion of average gonadal maturity stage (IV) was in December, January and February (Fig. 8). The data on maturity stages indicated that length at 50% maturity (L_{50}) were 8.94 cm TL for females (Fig. 9).

The sparse literature on silvery pout describes the species as bathypelagic (Sorbe 1980; Mauchline and Gordon 1984) with diet consisting of both pelagic components and benthopelagic or even epibenthic crustaceans (Macpherson 1981; Mattson 1981; Mauchline and Gordon 1984). It is further reported as an important prey for pelagic fish species, as blue whiting *Micromesistius poutassou* (Risso, 1827) and the demersal species cod *Gadus morhua* Linnaeus, 1758 and megrim *Lepidorhombus whiffiagonis* (Walbaum, 1792) (Armstrong 1982; Geistdoerfer 1983; Rodríguez-Cabello et al. 2014).

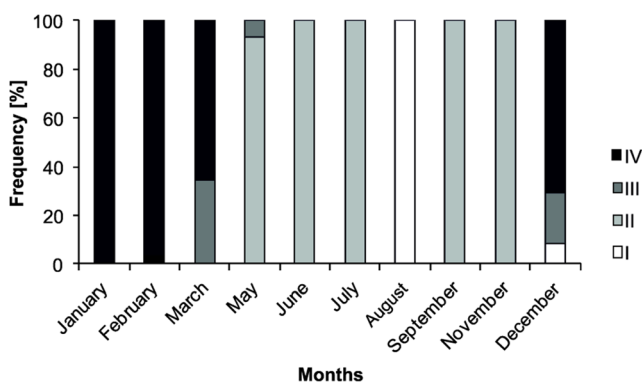


Fig. 8 Monthly variation of maturity stages for females, *G. argenteus* (I: immature; II: maturing; III: mature; IV: ripe)

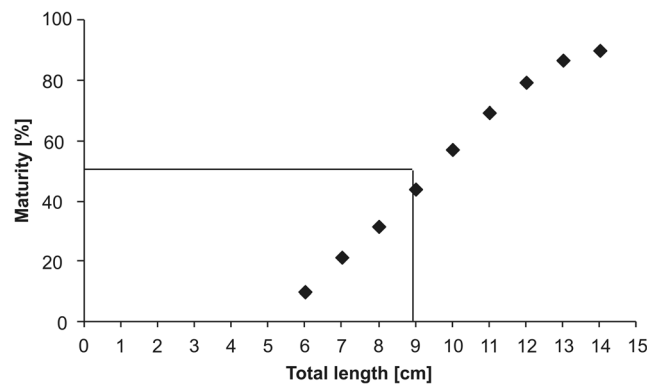


Fig. 9 Length at first maturity for females, *G. argenteus*

Most gadid species are economically important, although silvery pouts have limited commercial value due to their small size and predominantly artisanal use in some Mediterranean countries. This becomes evident considering the relatively limited number of studies present on these fishes compared to other, larger gadids (Gaemers and Poulsen 2017). Studies on life history parameters and fisheries of this species are generally insufficient. Studying this species is important to better understanding of its role in the deep sea ecosystem in the Saros Bay and at same time gain the knowledge on their age composition, growth and reproduction. Furthermore, this is the first report on the reproduction period and growth parameters of this species in the Aegean Sea. These information will help fisheries scientists in future studies of *G. argenteus* populations.

Acknowledgements The present study was financially supported of TUBITAK 106Y035. The authors would like to thank the crew of the trawl vessel ‘Şahin Reis’ and all the staff who helped in the fieldwork and in the laboratory.

Compliance with Ethical Standards

Conflict of Interest -On behalf of all authors, the corresponding author states that there is no conflict of interest.

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